

## PRDM14 promotes active DNA demethylation through the ten-eleven translocation (TET)-mediated base excision repair pathway in embryonic stem cells.

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Ten-eleven translocation (TET) proteins oxidize 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC). 5fC and 5caC can be excised and repaired by the base excision repair (BER) pathway, implicating 5mC oxidation in active DNA demethylation. Genome-wide DNA methylation is erased in the transition from metastable states to the ground state of embryonic stem cells (ESCs) and in migrating primordial germ cells (PGCs), although some resistant regions become demethylated only in gonadal PGCs. Understanding the mechanisms underlying global hypomethylation in naive ESCs and developing PGCs will be useful for realizing cellular pluripotency and totipotency. In this study, we found that PRDM14, the PR domain-containing transcriptional regulator, accelerates the TET-BER cycle, resulting in the promotion of active DNA demethylation in ESCs. Induction of Prdm14 expression transiently elevated 5hmC, followed by the reduction of 5mC at pluripotency-associated genes, germline-specific genes and imprinted loci, but not across the entire genome, which resembles the second wave of DNA demethylation observed in gonadal PGCs. PRDM14 physically interacts with TET1 and TET2 and enhances the recruitment of TET1 and TET2 at target loci. Knockdown of TET1 and TET2 impaired transcriptional regulation and DNA demethylation by PRDM14. The repression of the BER pathway by administration of pharmacological inhibitors of APE1 and PARP1 and the knockdown of thymine DNA glycosylase (TDG) also impaired DNA demethylation by PRDM14. Furthermore, DNA demethylation induced by PRDM14 takes place normally in the presence of aphidicolin, which is an inhibitor of G1/S progression. Together, our analysis provides mechanistic insight into DNA demethylation in naive pluripotent stem cells and developing PGCs.

## Minimal role of base excision repair in TET-induced global DNA demethylation in HEK293T cells.

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Oxidation of 5-methylcytosine by TET family proteins can induce DNA replication-dependent (passive) DNA demethylation and base excision repair (BER)-based (active) DNA demethylation. The balance of active vs. passive TET-induced demethylation remains incompletely determined. In the context of large scale DNA demethylation, active demethylation may require massive induction of the DNA repair machinery and thus compromise genome stability. To study this issue, we constructed a tetracycline-controlled TET-induced global DNA demethylation system in HEK293T cells. Upon TET overexpression, we observed induction of DNA damage and activation of a DNA damage response; however, BER genes are not upregulated to promote DNA repair. Depletion of TDG (thymine DNA glycosylase) or APEX1 (apurinic/apyrimidinic endonuclease 1), two key BER enzymes, enhances rather than impairs global DNA demethylation, which can be explained by stimulated proliferation. By contrast, growth arrest dramatically blocks TET-induced global DNA demethylation. Thus, in the context of TET-induction in HEK293T cells, the DNA replication-dependent passive mechanism functions as the predominant pathway for global DNA demethylation. In the same context, BER-based active demethylation is markedly restricted by limited BER upregulation, thus potentially preventing a disastrous DNA damage response to extensive active DNA demethylation.

## Emerging roles of TET proteins and 5-hydroxymethylcytosines in active DNA demethylation and beyond.

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Cytosine methylation is the major epigenetic modification of metazoan DNA. Although there is strong evidence that active DNA demethylation occurs in animal cells, the molecular details of this process are unknown. The recent discovery of the TET protein family (TET1-3) 5-methylcytosine hydroxylases has provided a new entry point to reveal the identity of the long-sought DNA demethylase. Here, we review the recent progress in understanding the function of TET proteins and 5-hydroxymethylcytosine (5hmC) through various biochemical and genomic approaches, the current evidence for a role of 5hmC as an early intermediate in active DNA demethylation and the potential functions of TET proteins and 5hmC beyond active DNA demethylation. We also discuss how future studies can extend our knowledge of this novel epigenetic modification.

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## Induced DNA demethylation by targeting Ten-Eleven Translocation 2 to the human ICAM-1 promoter.

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Increasing evidence indicates that active DNA demethylation is involved in several processes in mammals, resulting in developmental stage-specificity and cell lineage-specificity. The recently discovered Ten-Eleven Translocation (TET) dioxygenases are accepted to be involved in DNA demethylation by initiating 5-mC oxidation. Aberrant DNA methylation profiles are associated with many diseases. For example in cancer, hypermethylation results in silencing of tumor suppressor genes. Such silenced genes can be re-expressed by epigenetic drugs, but this approach has genome-wide effects. In this study, fusions of designer DNA binding domains to TET dioxygenase family members (TET1, -2 or -3) were engineered to target epigenetically silenced genes (ICAM-1, EpCAM). The effects on targeted CpGs' methylation and on expression levels of the target genes were assessed. The results indicated demethylation of targeted CpG sites in both promoters for targeted TET2 and to a lesser extent for TET1, but not for TET3. Interestingly, we observed re-activation of transcription of ICAM-1. Thus, our work suggests that we provided a mechanism to induce targeted DNA demethylation, which facilitates re-activation of expression of the target genes. Furthermore, this Epigenetic Editing approach is a powerful tool to investigate functions of epigenetic writers and erasers and to elucidate consequences of epigenetic marks.

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## Tet Proteins Regulate Neutrophil Granulation in Zebrafish through Demethylation of *socs3b* mRNA.

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Tet proteins (Tet1/2/3) convert 5-methylcytosine (5mC) to 5-hydroxy-methylcytosine (5hmC), initiating the process of active demethylation to regulate gene expression. Demethylation has been investigated primarily in the context of DNA, but recently Tet enzymes have also been shown to mediate demethylation of 5mC in RNA as an additional level of epitranscriptomic regulation. We analyzed compound tet2/3 mutant zebrafish and discovered a role for Tet enzymes in the maturation of primitive

and definitive neutrophils during granulation. Transcript profiling showed dysregulation of cytokine signaling in tet mutant neutrophils, including upregulation of *socs3b*. We show that Tet normally demethylates *socs3b* mRNA during granulation, thereby destabilizing the transcript, leading to its downregulation. Failure of this process leads to accumulation of *socs3b* mRNA and repression of cytokine signaling at this crucial step of neutrophil maturation. This study provides further evidence for Tets as epitranscriptomic regulatory enzymes and implicates Tet2/3 in regulation of neutrophil maturation.

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## Roles of TET and TDG in DNA demethylation in proliferating and non-proliferating immune cells.

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TET enzymes mediate DNA demethylation by oxidizing 5-methylcytosine (5mC) in DNA to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and 5-carboxylcytosine (5caC). Since these oxidized methylcytosines (oxi-mCs) are not recognized by the maintenance methyltransferase DNMT1, DNA demethylation can occur through "passive," replication-dependent dilution when cells divide. A distinct, replication-independent ("active") mechanism of DNA demethylation involves excision of 5fC and 5caC by the DNA repair enzyme thymine DNA glycosylase (TDG), followed by base excision repair. Here by analyzing inducible gene-disrupted mice, we show that DNA demethylation during primary T cell differentiation occurs mainly through passive replication-dependent dilution of all three oxi-mCs, with only a negligible contribution from TDG. In addition, by pyridine borane sequencing (PB-seq), a simple recently developed method that directly maps 5fC/5caC at single-base resolution, we detect the accumulation of 5fC/5caC in TDG-deleted T cells. We also quantify the occurrence of concordant demethylation within and near enhancer regions in the *Il4* locus. In an independent system that does not involve cell division, macrophages treated with liposaccharide accumulate 5hmC at enhancers and show altered gene expression without DNA demethylation; loss of TET enzymes disrupts gene expression, but loss of TDG has no effect. We also observe that mice with long-term (1 year) deletion of *Tdg* are healthy and show normal survival and hematopoiesis. We have quantified the relative contributions of TET and TDG to cell differentiation and DNA demethylation at representative loci in proliferating T cells. We find that TET enzymes regulate T cell differentiation and DNA demethylation primarily through passive dilution of oxi-mCs. In contrast, while we observe a low level of active, replication-independent DNA demethylation mediated by TDG, this process does not appear to be essential for immune cell activation or differentiation.

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## Concomitant loss of TET2 and TET3 results in T cell expansion and genomic instability in mice.

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Ten eleven translocation (TET) proteins are tumor suppressors that through their catalytic activity oxidize 5-methylcytosine to 5-hydroxymethylcytosine, to promote DNA demethylation and to regulate gene expression. Notably, TET2 is one of the most frequently mutated genes in hematological malignancies, including T cell lymphomas. However, murine models with deletion of TET2 do not exhibit T cell expansion, presumably due to redundancy with other members of the TET family of proteins. In order to gain insight on the TET mediated molecular events that safeguard T cells from aberrant proliferation we

performed serial adoptive transfers of murine CD4 T cells that lack concomitantly TET2 and TET3 to fully immunocompetent congenic mice. Here we show a progressive acquisition of malignant traits upon loss of TET2 and TET3 that is characterized by loss of genomic integrity, acquisition of aneuploidy and upregulation of the protooncogene Myc.

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## DNA methylation modification in Idiopathic pulmonary fibrosis.

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Idiopathic pulmonary fibrosis (IPF) is a chronic, progressive, and irreversible interstitial lung disease with a prognosis worse than lung cancer. It is a fatal lung disease with largely unknown etiology and pathogenesis, and no effective therapeutic drugs render its treatment largely unsuccessful. With continuous in-depth research efforts, the epigenetic mechanisms in IPF pathogenesis have been further discovered and concerned. As a widely studied mechanism of epigenetic modification, DNA methylation is primarily facilitated by DNA methyltransferases (DNMTs), resulting in the addition of a methyl group to the fifth carbon position of the cytosine base, leading to the formation of 5-methylcytosine (5-mC). Dysregulation of DNA methylation is intricately associated with the advancement of respiratory disorders. Recently, the role of DNA methylation in IPF pathogenesis has also received considerable attention. DNA methylation patterns include methylation modification and demethylation modification and regulate a range of essential biological functions through gene expression regulation. The Ten-Eleven-Translocation (TET) family of DNA dioxygenases is crucial in facilitating active DNA demethylation through the enzymatic conversion of the modified genomic base 5-mC to 5-hydroxymethylcytosine (5-hmC). TET2, a member of TET proteins, is involved in lung inflammation, and its protein expression is downregulated in the lungs and alveolar epithelial type II cells of IPF patients. This review summarizes the current knowledge of pathologic features and DNA methylation mechanisms of pulmonary fibrosis, focusing on the critical roles of abnormal DNA methylation patterns, DNMTs, and TET proteins in impacting IPF pathogenesis. Researching DNA methylation will enhance comprehension of the fundamental mechanisms involved in IPF pathology and provide novel diagnostic biomarkers and therapeutic targets for pulmonary fibrosis based on the studies involving epigenetic mechanisms.

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## Loss of Tet hydroxymethylase activity causes mouse embryonic stem cell differentiation bias and developmental defects.

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The TET family is well known for active DNA demethylation and plays important roles in regulating transcription, the epigenome and development. Nevertheless, previous studies using knockdown (KD) or knockout (KO) models to investigate the function of TET have faced challenges in distinguishing its enzymatic and nonenzymatic roles, as well as compensatory effects among TET family members, which has made the understanding of the enzymatic role of TET not accurate enough. To solve this problem, we successfully generated mice catalytically inactive for specific Tet members (Tetm/m). We observed that, compared with the reported KO mice, mutant mice exhibited distinct developmental defects, including growth retardation, sex imbalance, infertility, and perinatal lethality. Notably, Tetm/m mouse embryonic stem cells (mESCs) were successfully established but entered an impaired developmental program, demonstrating extended pluripotency and defects in ectodermal differentiation caused by abnormal DNA

methylation. Intriguingly, Tet3, traditionally considered less critical for mESCs due to its lower expression level, had a significant impact on the global hydroxymethylation, gene expression, and differentiation potential of mESCs. Notably, there were common regulatory regions between Tet1 and Tet3 in pluripotency regulation. In summary, our study provides a more accurate reference for the functional mechanism of Tet hydroxymethylase activity in mouse development and ESC pluripotency regulation.

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## Perturbing TET2 condensation promotes aberrant genome-wide DNA methylation and curtails leukaemia cell growth.

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The ten-eleven translocation (TET) family of dioxygenases maintain stable local DNA demethylation during cell division and lineage specification. As the major catalytic product of TET enzymes, 5-hydroxymethylcytosine is selectively enriched at specific genomic regions, such as enhancers, in a tissue-dependent manner. However, the mechanisms underlying this selectivity remain unresolved. Here we unveil a low-complexity insert domain within TET2 that facilitates its biomolecular condensation with epigenetic modulators, such as UTX and MLL4. This co-condensation fosters a permissive chromatin environment for precise DNA demethylation. Disrupting low-complexity insert-mediated condensation alters the genomic binding of TET2 to cause promiscuous DNA demethylation and genome reorganization. These changes influence the expression of key genes implicated in leukaemogenesis to curtail leukaemia cell proliferation. Collectively, this study establishes the pivotal role of TET2 condensation in orchestrating precise DNA demethylation and gene transcription to support tumour cell growth.

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